



Final year project 3A - IRIS

Exploring Exemplar-Based Continual Learning Strategies for Image Classification

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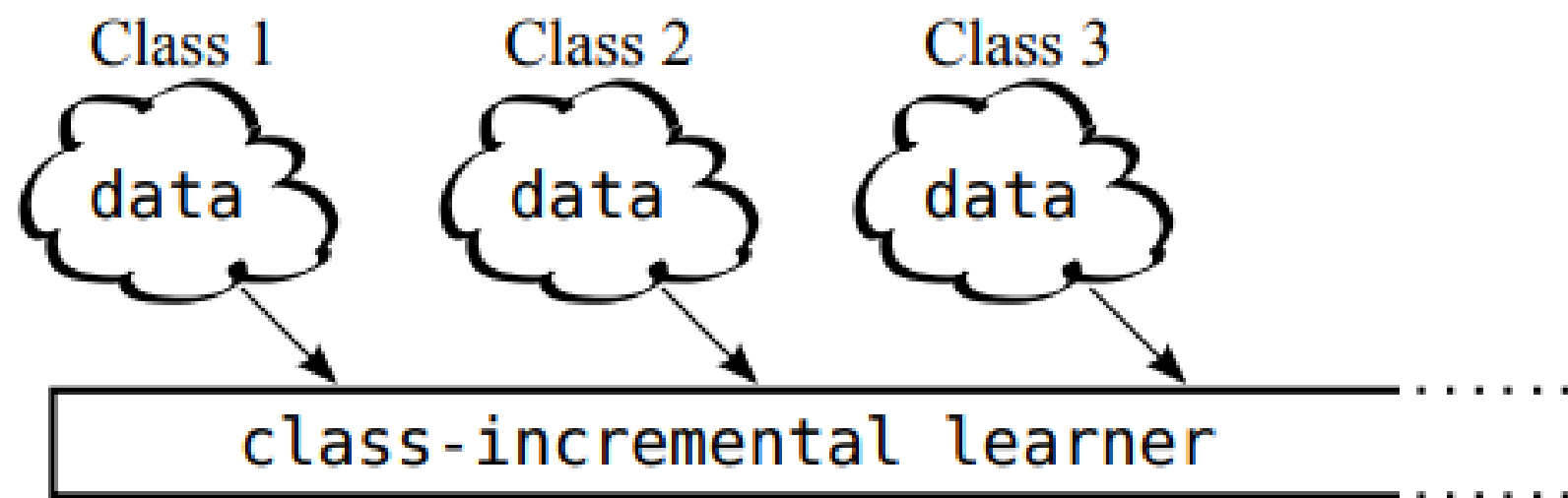
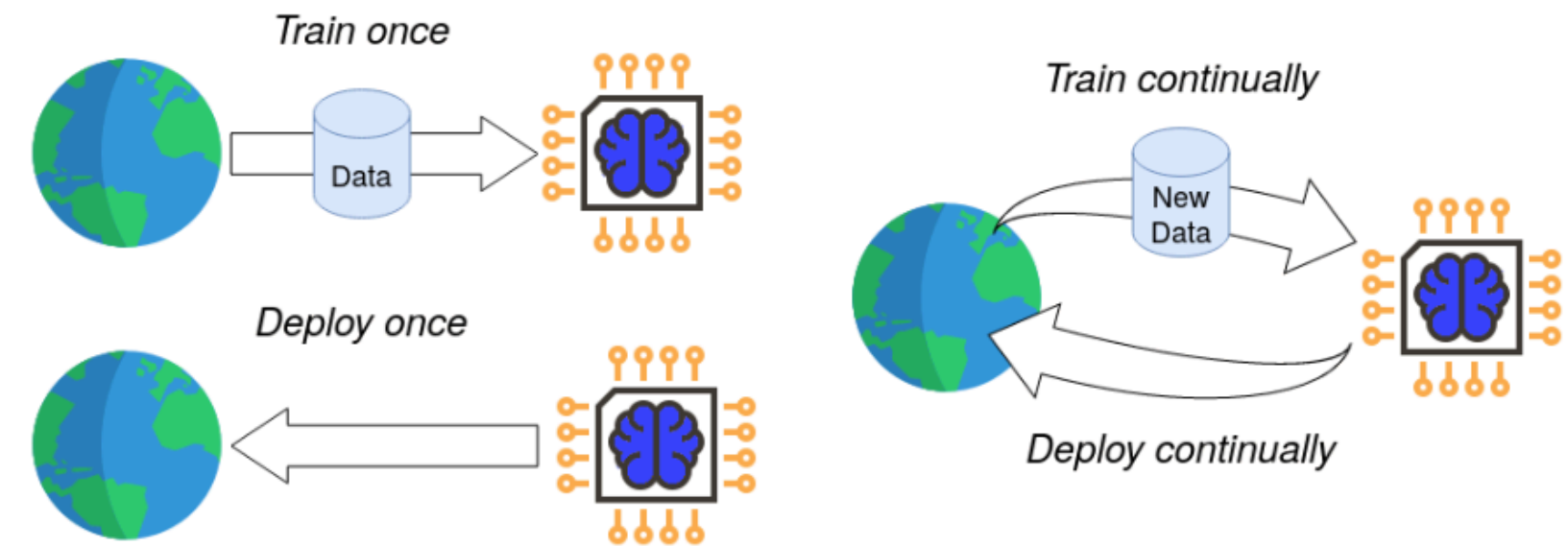
Supervised by:
Mr. Cong Tu TRAN

OUTLINE

- Problem definition
- iCaRL: Incremental Classifier and Representation Learning
- MIR: Maximally Interfered Retrieval
- Conclusion

Continual Learning (CL)

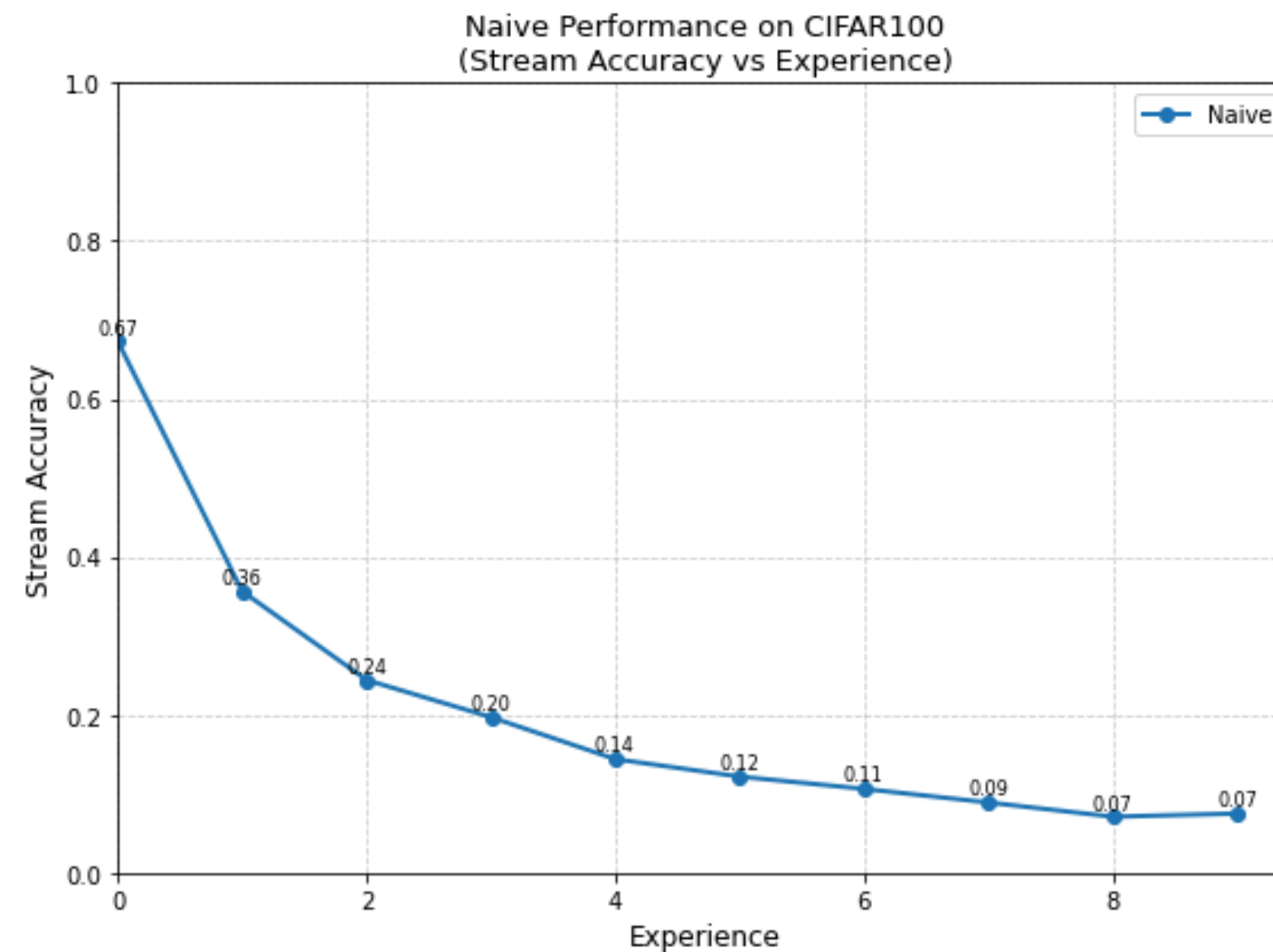
- Instance-Incremental Learning (IIL)
- Task-Incremental Learning (TIL)
- Domain-Incremental Learning (DIL)
- Class-Incremental Learning (CIL)**

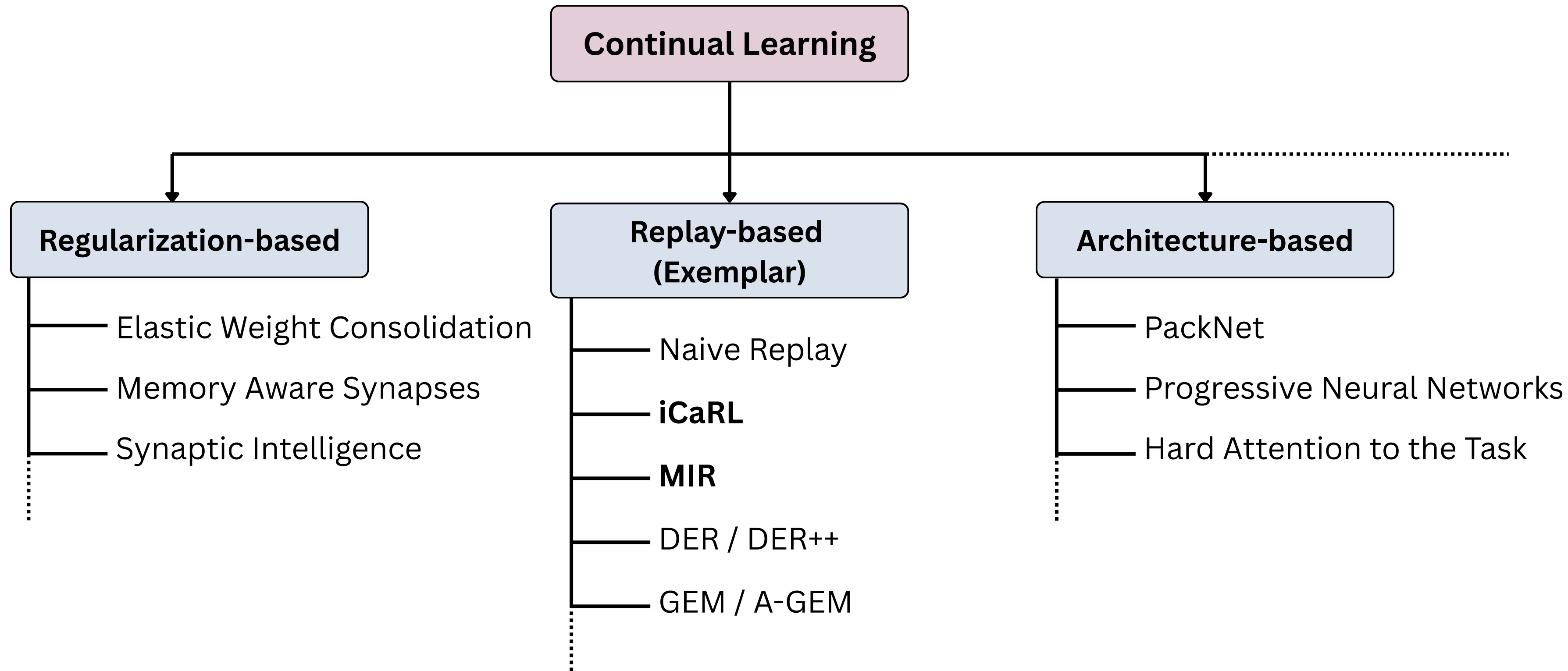


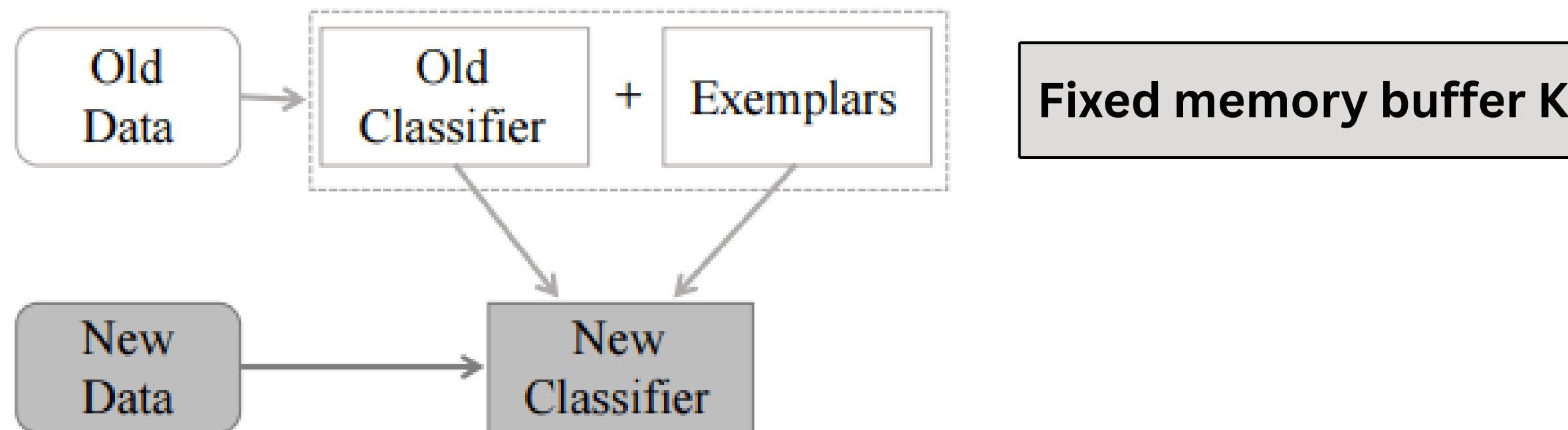
- Data is not available all at once.
- Learning happens experience after experience.
- The model must adapt to new tasks without retraining from scratch.

Catastrophic Forgetting

- The model learns a new task.
- Performance on previous tasks drops drastically.







Exemplar Selection (Herding)



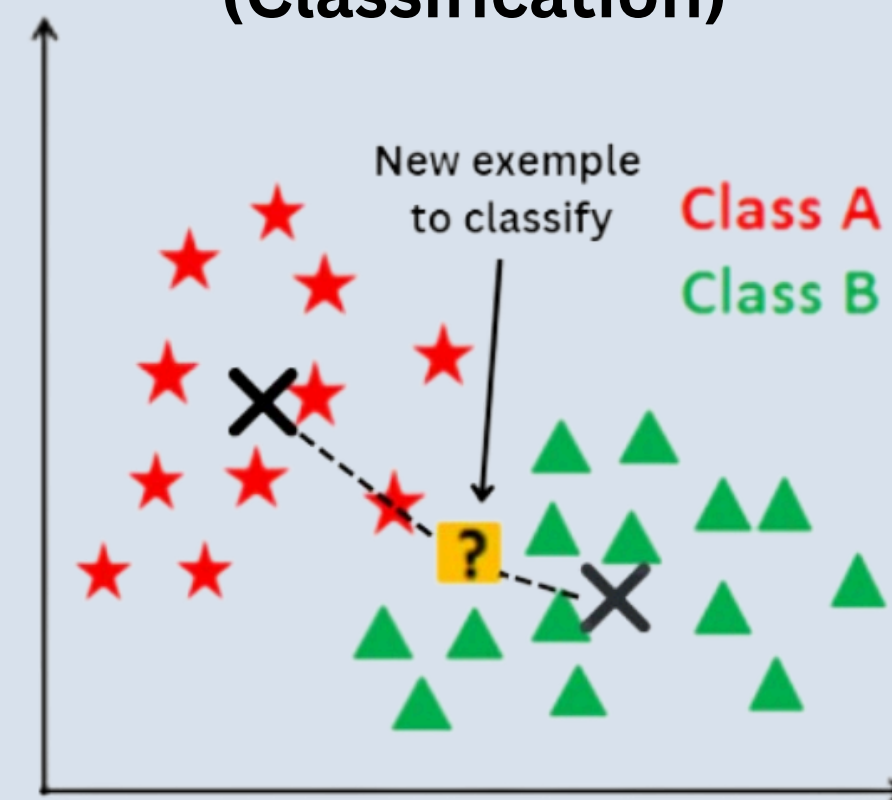
- Minimize distance to class mean
- “Most represented” samples

Knowledge Distillation (Loss)

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{dist}$$

- Prevent forgetting old knowledge during new training

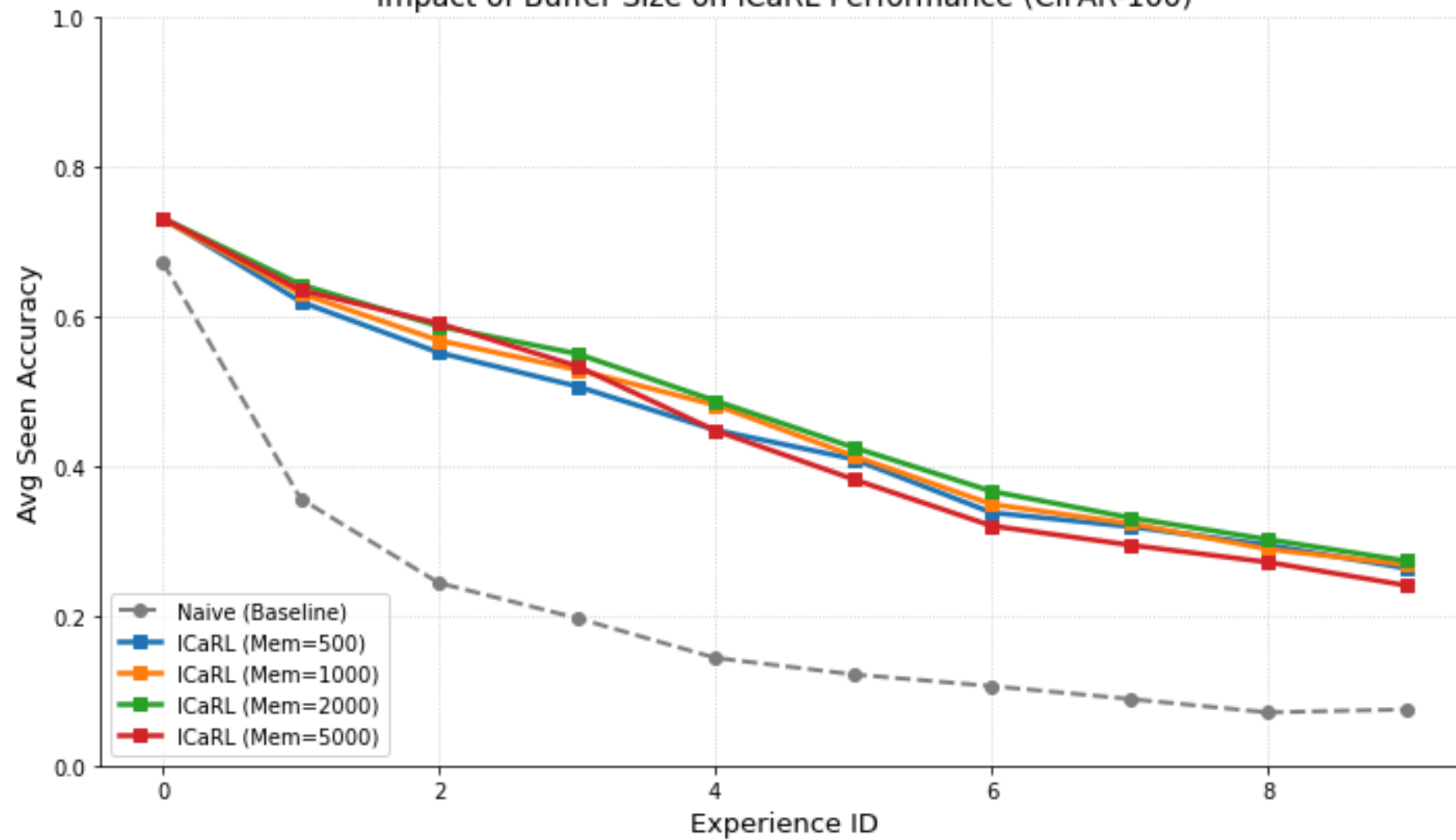
Nearest-Mean-of-Exemplars (Classification)



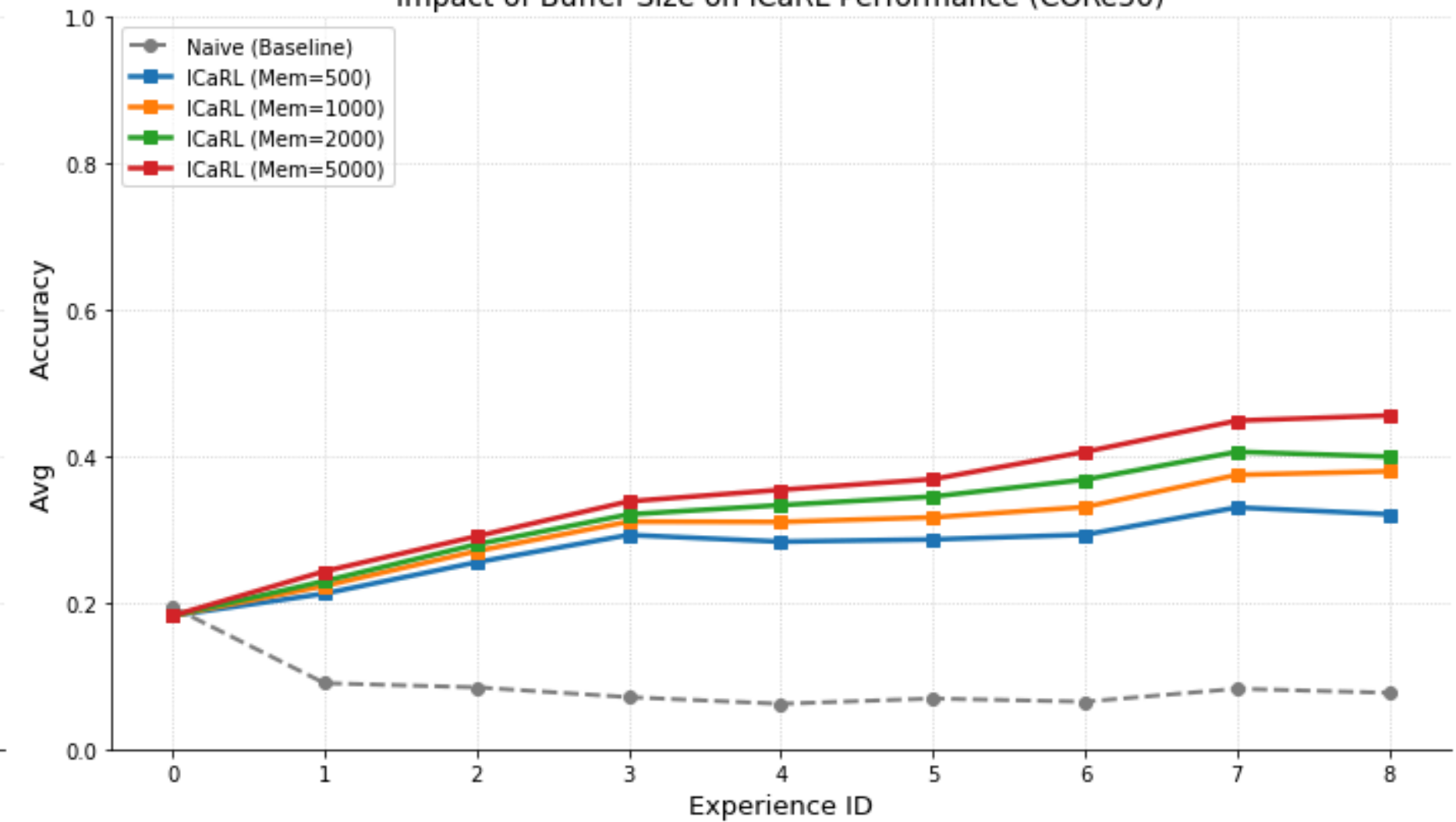
- Classification based on similarity to class means

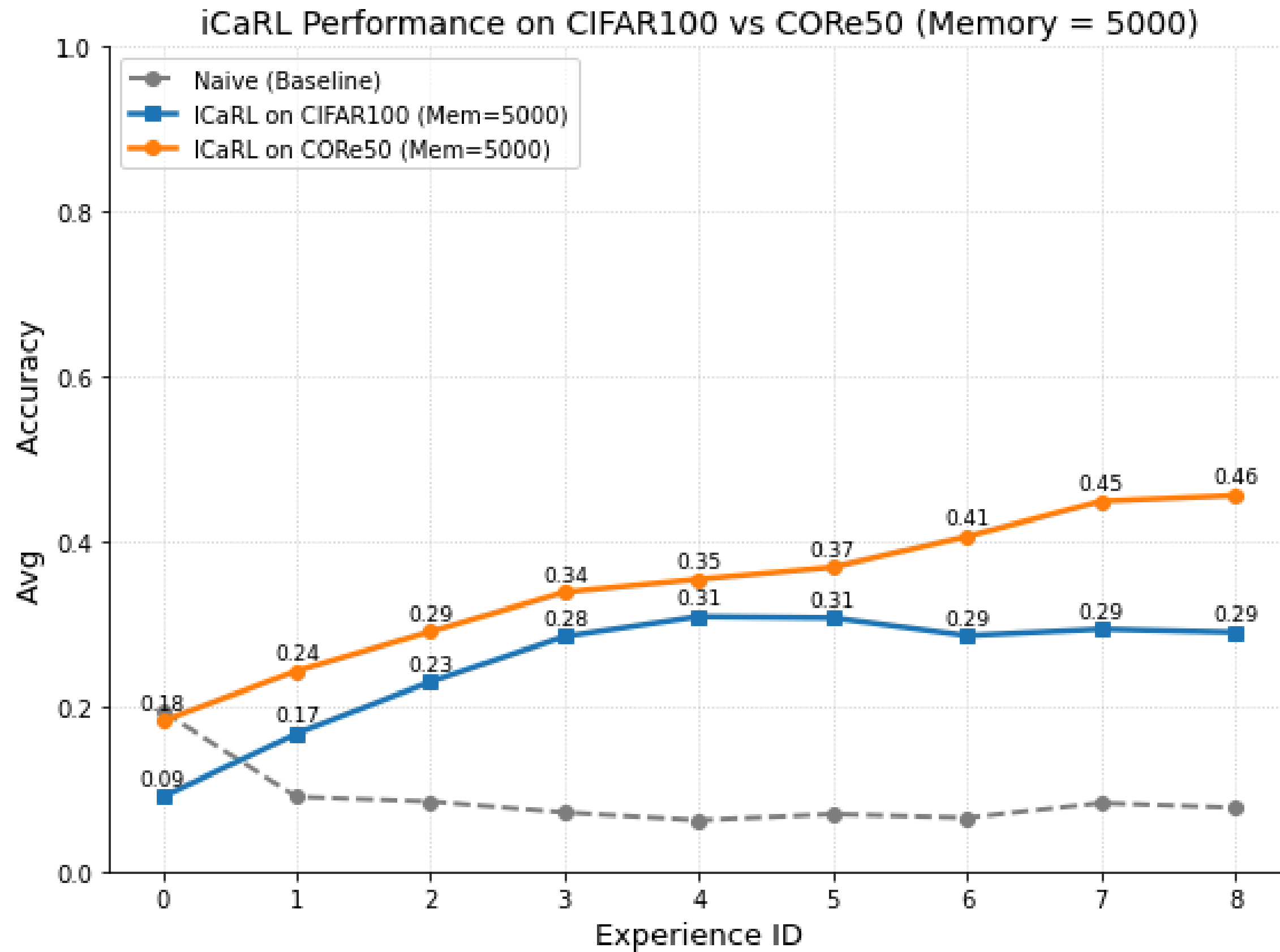
Feature	Split CIFAR-100	CORe50 (New Classes)
Domain	Natural Images (Static)	Robotics / Real-world Objects
Scale	100 Classes (32x32)	50 Objects (128x128)
Experiences	10 (10 new classes each)	9 (10 then 5 each)
Total Images	60,000	~164,000

Impact of Buffer Size on iCaRL Performance (CIFAR-100)

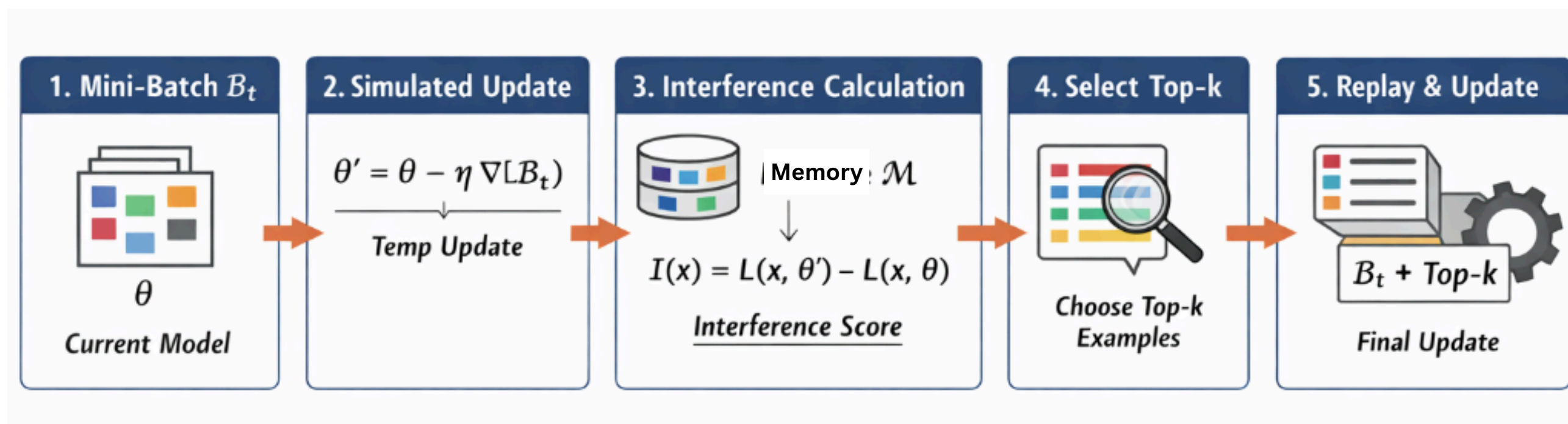


Impact of Buffer Size on iCaRL Performance (CORe50)

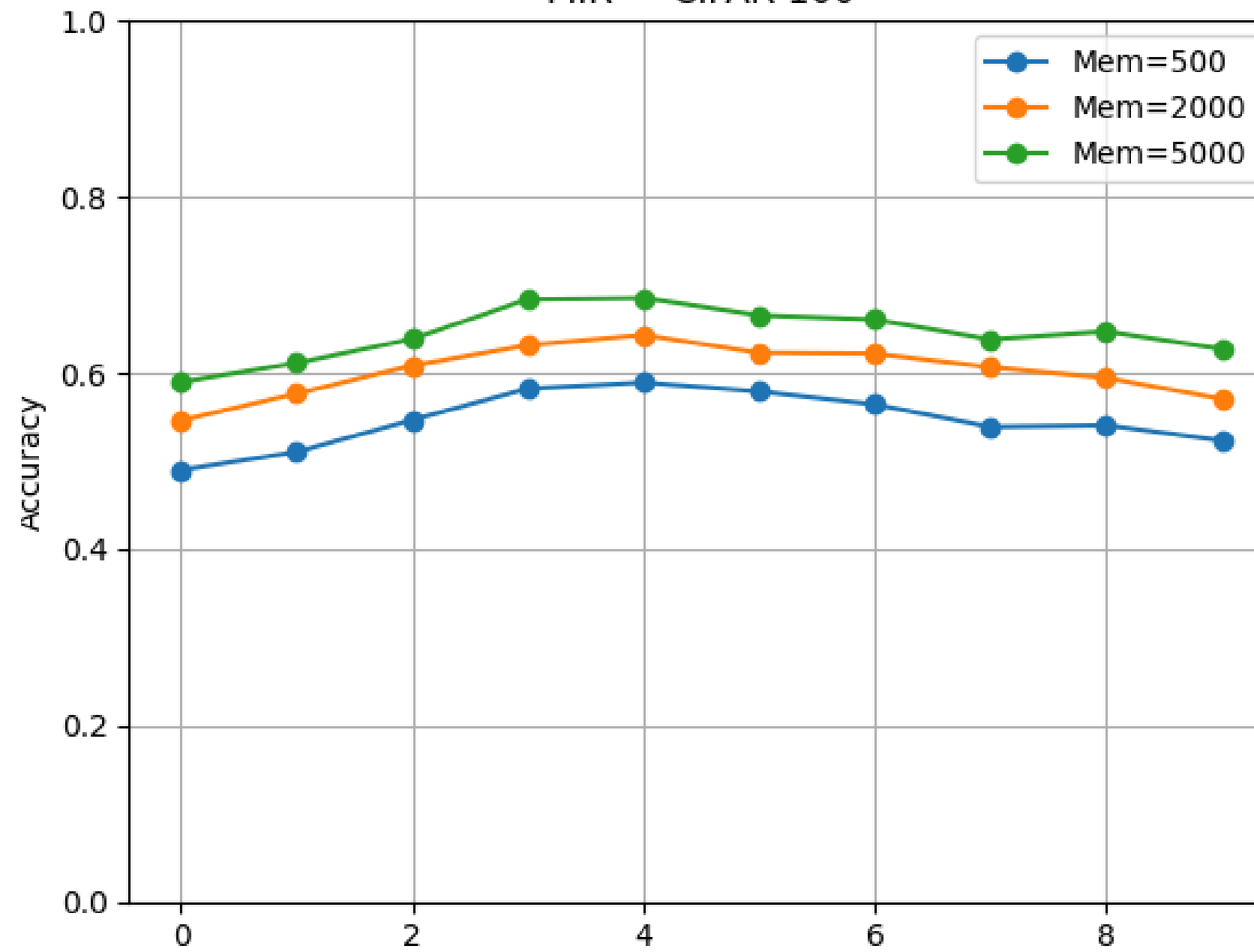




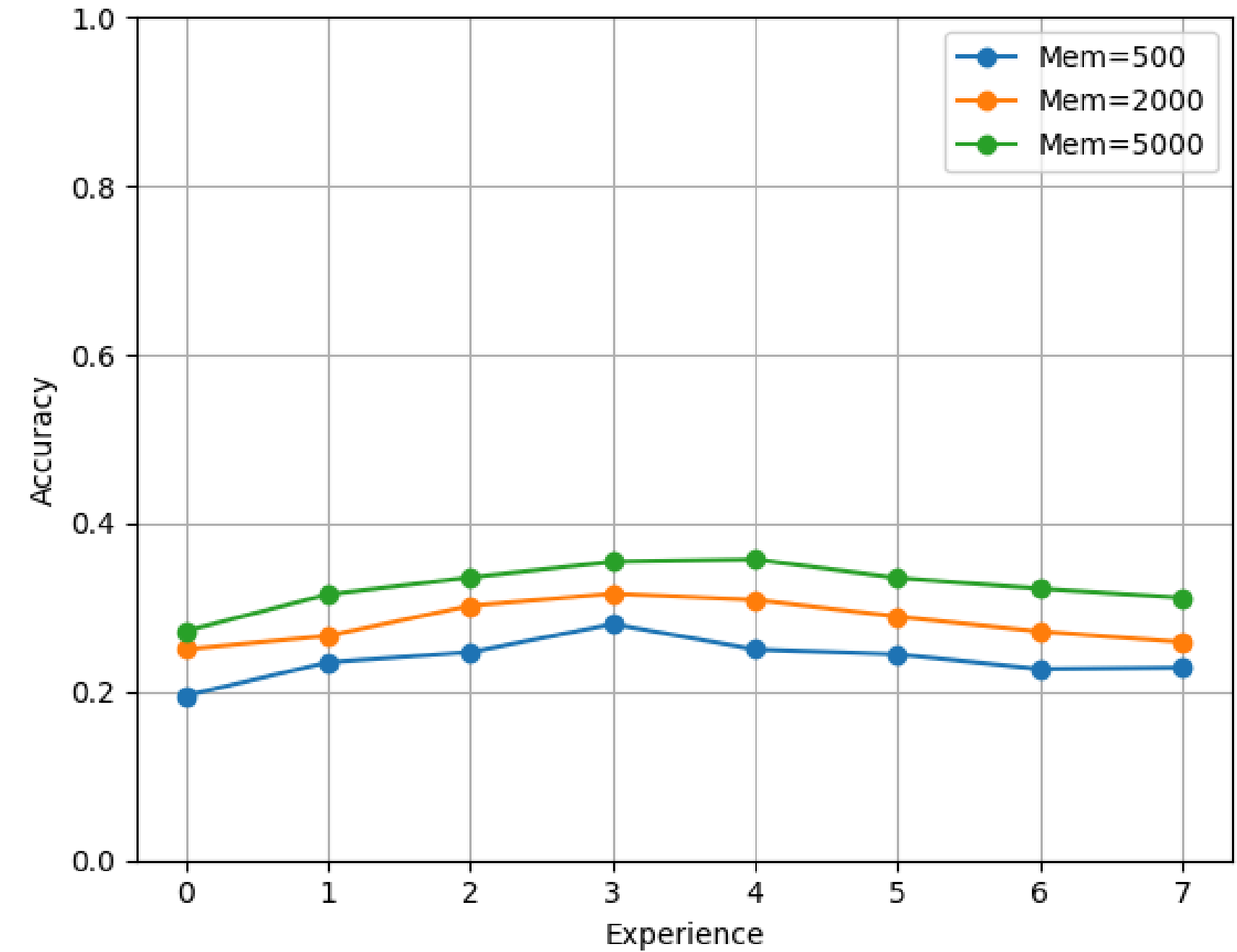
- Instead of random replay → selects most interfered samples.
- Focuses on samples most at risk of being forgotten
- Replay intelligent



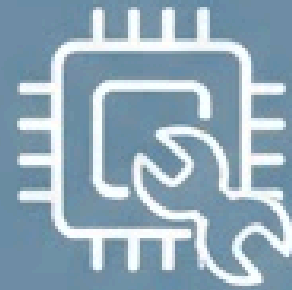
MIR — CIFAR-100



MIR — CORe50



Challenges & Difficulties



Computational
Constraints

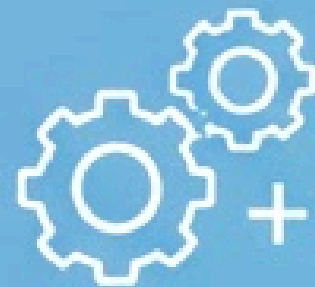


Hyperparameter Tuning



Framework Complexity

Future Perspectives



Hybrid Architectures



Self-Supervised
Features

THANK YOU!